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Modification of network structure induced by glass former composition and its correlation to the conductivity in lithium borophosphate glass for solid state electrolyte

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Abstract

The $x\mathrm{Li_2O}$–$(1-x)(y\mathrm{B_2O_3}$–$(1-y)\mathrm{P_2O_5})$ glasses system is one of the parent compositions for chemically and electrochemically stable solid state electrolyte applicable to thin film battery. The purpose of this study was to seek an optimum composition maximizing the ionic conductivity for the deposition of thin film electrolytes. $x\mathrm{Li_2O}$–$(1-x)(y\mathrm{B_2O_3}$–$(1-y)\mathrm{P_2O_5})$ glasses were prepared with wide range of composition, i.e. x = 0.40–0.47 and $B_2O_3/(B_2O_3+P_2O_5)$ ratio was 0.17–0.67 (y = xx ~ xx). The structural variation of the oxide Li_2O–B_2O_3–P_2O_5 system was investigated by Raman spectroscopy to find out the correlation with the electrical conductivity. It was shown that under 40 mol% of Li_2O the ionic conductivity monotonically increased with increasing of B_2O_3, while over the 40 mol% of Li_2O it exhibited a maximum at a specific $B_2O_3/(B_2O_3+P_2O_5)$ ratio, typically ~ 0.5. The structural investigation indicated that the variation in conductivity was closely correlated to the variation of 4-coordinated boron groups.

Keywords: Solid state electrolyte; Glass electrolyte; Conductivity

1. Introduction

During the last two decades, inorganic glasses have long been studied as a candidate for solid state electrolyte material of the rechargeable thin film battery which will be applied to electronic devices such as micro-electro-mechanical system, smart card, medical appliance, and so on. However, solid state lithium batteries have not been widely commercialized since solid electrolytes developed so far have not completely satisfied the requirements for practical applications, where high ionic conductivity as well as chemical, thermal and electrochemical stability are required [1].

Among solid electrolytes being considered, inorganic solid electrolytes have drawn less attention since they are impractical to apply to common bulk type batteries. However, when considering the thin film type battery, inorganic solid electrolytes become a first choice since various deposition techniques for these materials have already been demonstrated successfully. In thin film approach, the fatal drawback of inorganic solid electrolyte, low lithium-ion conductivity, can be overcome by reducing the thickness of electrolyte. The inorganic electrolytes reported so far are classified into amorphous electrolytes and crystalline electrolytes, and the inorganic glass electrolytes became a practical candidate due to several advantages such as the ease of deposition and wide choice of material system and composition range as well as the electrochemical stability and the chemical inertness against the active materials employed lithium battery. Among lithium-ion conductive glasses [2], sulfide glasses are limited in practical application due to its instability to moisture in atmosphere and lithium metal [3] in spite of their high ionic conductivity of $10^{-3} \sim 10^{-4}$ S/cm at room temperature. The difficulty in handling in normal deposition apparatus is another obstacle for the thin film process. On the other hand, oxide glasses are generally chemically and electrochemically more stable than sulfide glasses, and it is relatively easier to fabricate into thin film type electrolyte for microbattery applications. Therefore, it is believed that glass electrolyte can be a strong candidate for

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the solid state thin film battery. Furthermore, its viscous flow property at elevated temperature enables a 3 dimensional contact to porous active materials ensuring enough electrochemical reactions of lithium ion at the interfaces.

Although the thin film scheme of battery cell reduces the limitation of inorganic glasses, it is certain that low lithium-ion conductivity of oxide glasses, $10^{7}\sim10^{-8}$ S/cm at room temperature, should be improved. Thus, the major effort has been focused on the improvement of the ion conductivity of oxide glasses [3,4].

In the present work, the lithium borophosphate glass system was selected as a candidate for thin film electrolyte since it offers the freedom in deposition technique, the composition of choice, material stability, high temperature viscous flow properties, and the resistance to devitrification. Prior to the fabrication of thin film electrolyte, the ionic conductivity of the lithium borophosphate glass system was studied in full range of composition of glass former and modifier to prepare the basic engineering data. The structural variation of the oxide amorphous electrolytes of Li_2O–B_2O_3–P_2O_5 system was investigated by Raman spectroscopy to find out the correlation with the electrical conductivity.

2. Experiments

Glass electrolytes with composition $x$$Li_2O$–$(1-x)(yB_2O_3$–$(1-y)P_2O_5)$ were prepared from reagent grade material (99% Li_2CO_3, 98% P_2O_5 and 95% B_2O_3). The starting materials were mixed in proportions appropriate to form 20 g batches. The homogeneously mixed powders were then melted in Pt crucibles at 1000 °C for 5 h using an electric furnace and air-quenched in a stainless-steel mold to give transparent glass plates of 1 mm–2 mm thick. The amorphous state of glass electrolytes was confirmed at room temperature by X-ray diffraction (XRD). Then, for lithium-ion conductivity measurement, Pt was sputtered on both sides of the polished specimen as ion blocking electrodes. The conductivity measurements were performed by complex impedance method using Solartron impedance analyzer with an ac voltage of 1 V amplitude over the frequency range 10 Hz to 1 MHz at room temperature. The correlation between the structural modification of glass network and conductivity was characterized by Raman spectroscopy as a function of the chemical composition in the range from 400 to 4000 cm^{-1}. The Raman spectra of the glasses were performed using a laser Raman Spectrophotometer (JASCO NRS-3100) and a conventional counting system in 400–1800 cm^{-1} range in the air.

3. Results and discussion

Glass forming region in the system of xLi_2O–$(1-x)(yB_2O_3$–$(1-y)P_2O_5)$ was $x=0.40\sim0.47$ at the $B_2O_3/(B_2O_3+P_2O_5)=0.33$. The devitrification was checked by X-ray diffraction patterns shown in Fig. 1, which confirms that all of glass the specimens were quenched into amorphous glass.

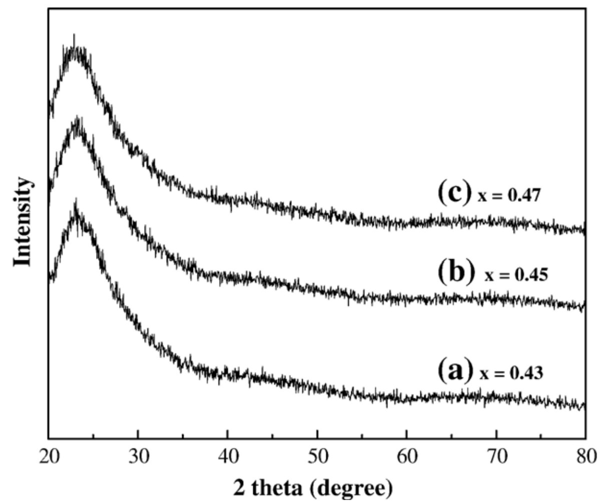

Fig. 1. X-ray diffraction pattern of the prepared glass electrolytes with the compositions of xLi_2O–$(1-x)(yB_2O_3$–$(1-y)P_2O_5)$ at various Li_2O composition, x. $B_2O_3/(B_2O_3+P_2O_5)$ ratio was 0.33.

Fig. 2, the typical examples of complex impedance plot of the xLi_2O–$(1-x)(yB_2O_3$–$(1-y)P_2O_5)$ at $B_2O_3/(B_2O_3+P_2O_5)=0.5$, presents the conductivity variation with mole fraction of network modifier, Li_2O. As shown in Fig. 2, the lithium-ion conductivity of glasses increased as increasing network modifier contents up to $x=0.45$, followed by abrupt decrease at $x=47$ mol%. The ionic conductivities of glasses are summarized in Fig. 3 as a function of $B_2O_3/(B_2O_3+P_2O_5)$ at various concentration of lithium oxide added to glass. One can see that, for $x\geq0.43$ under fixed Li_2O concentration, the ionic conductivity generally increased with increasing $B_2O_3/(B_2O_3+P_2O_5)$ ratio below $B_2O_3/(B_2O_3+P_2O_5)=0.5$ and decreased precipitously over $B_2O_3/(B_2O_3+P_2O_5)=0.5$. In the other region, when $B_2O_3/(B_2O_3+P_2O_5)$ ratio >0.5, the amorphous electrolyte started to exhibit a sign of local ordering in atomic scale and, at $Li_2O=47$ mol%, a partially ordered structure of the glass was observed. By this reason, the conductivity decreased precipitously at $B_2O_3/(B_2O_3+P_2O_5)$ ratio=0.67 and the correlation with structural change to the ionic conductivity will be discussed in the later section. Ionic conductivity is described by $\sigma=\sum n_i z_i e\mu_i$ (where n_i is the number of mobile ions, z_i is the valence of carrier ions, e is electric charge and μ_I is ionic mobility). Here, the ionic conductivity increased with increasing Li_2O content due to the increase of both the mobility and number of charge carrier. The increase of charge carrier number is not linearly proportional to the Li_2O content since the oxide glasses with alkali ions are generally weak electrolytes [5] or the lithium ions in the site of deep potential well are immobile. The estimation of the ionization fraction of added lithium is not a routine work and the method for indirect determination has been suggested previously [6]. The increase in lithium-ion mobility is due to the formation of the non-bridging oxygens or broken bonds within glass network. Non-bridging oxygen site is known to offer the hoping site for ionic conduction in oxide glass network where cation such as Li^+ jumps into or out easily due to relatively weak bonding or shallow energy well [7]. The formation of non-bridging oxygen is also contributing to the relatively open network structure with large free volume for ion

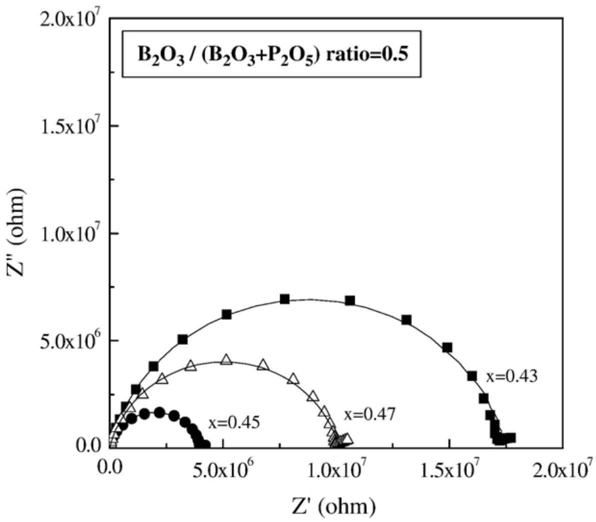

Fig. 2. Complex impedance plots of xLi$_2$O–(1−x)(yB$_2$O$_3$–(1−y)P$_2$O$_5$) glass electrolyte (amplitude: 1 V, frequency range:10 Hz–1 MHz, at 303 k).

drift. The combination of these two effects increases the ionic mobility of glass electrolyte.

It is generally known that 4-coordinated boron atoms in a glass network are more contributing to the ionic conduction [8–12] since its large molecular diameter reduces the ionic field strength and Li^+ ions are weakly bounded compared to non-bridging oxygen in 3-coordinated boron structure. Therefore, to find out the correlation between the structure of glass network and ionic conductivity, it is essential to watch the variation of 4-coordinated boron with non-bridging oxygen. In overall, the RAMAN spectrum of lithium borophosphate glass shows the characteristic peaks located around 1110–1080 cm^{-1}, 1030–1010 cm^{-1}, 770–730 cm^{-1}, and 670–630 cm^{-1} bands [13]. It is certain that 650 cm^{-1} band is closely related to the presence of $(BO_4+(PO_3)_n^{-n})$ and $(BO_2)^-$units. The peak 730 cm^{-1} is assigned to the vibration of diborate groups $(B_4O_7)^{-2}$, which is a combination of trigonal boron and tetragonal boron [14]. The 1025 cm^{-1} band is attributed to the overlapping vibrations of

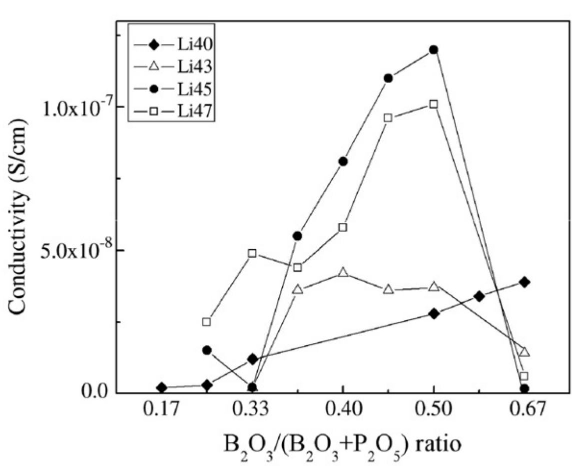

Fig. 3. The conductivity of the glass specimen as functions of B$_2$O$_3$/(B$_2$O$_3$+P$_2$O$_5$) ratio at various Li$_2$O concentration (DC conductivity, at room temperature).

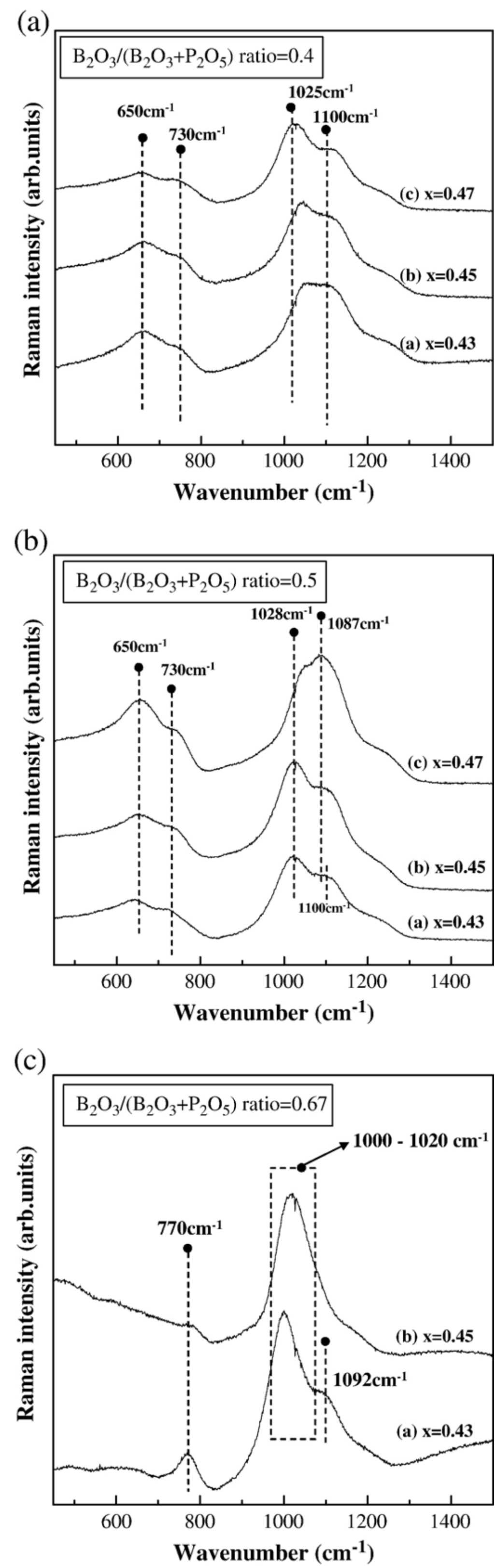

Fig. 4. The variation of RAMAN spectra of xLi$_2$O–(1−x)(yB$_2$O$_3$–(1−y)P$_2$O$_5$) glasses with various Li$_2$O, x, and B$_2$O$_3$/(B$_2$O$_3$+P$_2$O$_5$) ratio or y. (a) B$_2$O$_3$/(B$_2$O$_3$+P$_2$O$_5$)=0.4, (b) B$_2$O$_3$/(B$_2$O$_3$+P$_2$O$_5$)=0.5, and (c) B$_2$O$_3$/(B$_2$O$_3$+P$_2$O$_5$)=0.67.

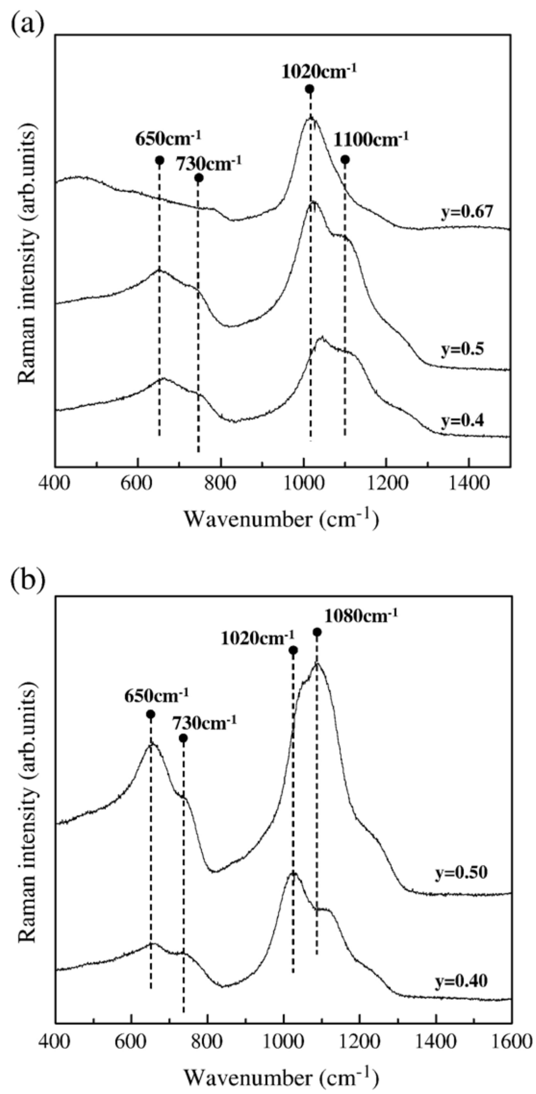

Fig. 5. The variation of RAMAN spectra of xLi$_2$O–(1−x)(yB$_2$O$_3$–(1−y)P$_2$O$_5$) glasses. y in figure is B$_2$O$_3$/(B$_2$O$_3$+P$_2$O$_5$) ratio. (a) x=0.45 and y=0.40–0.67 and (b) x=0.47 and y=0.40–0.50.

$(P_2O_7)^{-4}$ and (B–O–P)$^-$ groups, and the 1110 cm^{-1} band to the overlapping vibration of $(P_4O_{13})^{-6}$, $(P_3O_{10})^{-5}$ and (B–O–P)^0 group [15]. The number of 4-coordinated boron is closely related to the intensity of the peaks at 1080–1010 cm^{-1} and 670–630 cm^{-1}. The ionic conductivity conforms to the variation of these peaks.

Fig. 4 presents the variation of RAMAN spectra as a function of mole fraction of network modifier Li_2O with various fixed $B_2O_3/(B_2O_3+P_2O_5)$ ratio. For low $B_2O_3/(B_2O_3+P_2O_5)$ ratio (Fig. 4(a)), the peak at 650 and 1025 cm^{-1} increased with increasing modifier concentration, indicating that the ionic conductivity increases due to increasing the number of 4-coordinated boron[16,17]. This tendency was maintained up to $B_2O_3/(B_2O_3+P_2O_5)$ ratio=0.5 except for the case of x=0.47. For x=0.47, the conductivity was lower than that of x=0.45 in the range of $B_2O_3/(B_2O_3+P_2O_5)$ ratio=0.35 ~ 0.5. The reason is clear if one notices the variation of the 1028 cm^{-1} peak as shown in Fig. 4(b). The 1028 cm^{-1} peak reduced when x increased from 0.45 to 0.47 while 1087 cm^{-1} peak grew, suggesting that the 4-coordinated boron with weakly bonded Li^+ ion is converted into the larger clusters of 3-coordinated boron or P–O chains with non-bridging oxygen. This structural variation accompanies the escape of Li^+ ion from B–O chains and the preferential segregation into P–O chains. As mentioned earlier, this variation reduces the ionic mobility of Li^+ ion. This tendency becomes more clear when the $B_2O_3/(B_2O_3+P_2O_5)$ ratio increased over 0.5. For $B_2O_3/(B_2O_3+P_2O_5)$=0.67 (Fig. 4(c)), the Raman peaks became sharper indicating that the glass network became more “ordered”. Unlike to Fig. 4(b), 1020 cm^{-1} peak and 1090 cm^{-1} peak seem to be no longer separable at x=0.45, indicating that the local ordering undergoes in atomic scale. The tendency of local ordering coincides with the experimental observations. For x=0.47, it was hard to obtain the amorphous glass. Therefore, it is ambiguous to judge the structural variation from 1020 cm^{-1} peak and 1090 cm^{-1} peak at this case. However, one can see that the 770 cm^{-1} peak almost disappeared for x=0.45 and this suggests that BO_4 tetrahedral groups almost disappeared and this leads to the appreciable reduction in conductivity. This observation explains the abrupt decrease in conductivity at $B_2O_3/(B_2O_3+P_2O_5)$=0.67.

It is obvious that in lithium borophosphate glass, when $B_2O_3/(B_2O_3+P_2O_5)$ ratio ≤0.5, the number of tetrahedral boron is increasing with the increasing Li_2O concentration and, as a result, the ionic conductivity increases. On the other hand, when $B_2O_3/(B_2O_3+P_2O_5)$ ratio >0.5, the structural modification represented by “ordering” and “preferential segregation” of Li^+ ion into P–O chains is enhanced by the addition of Li_2O and this reduces the ionic conductivity. This structural modification can be seen more clearly in Fig. 5.

4. Conclusions

Solid state glass electrolytes with the composition xLi$_2$O–(1−x)(yB$_2$O$_3$–(1−y)P$_2$O$_5$) were prepared by melt quenching method. P_2O_5 was added into binary Li_2O–B_2O_3 glass electrolyte as a network former to prevent the devitrification and help the stable formation of amorphous phase of the glass electrolyte. In this system, 0.45Li$_2$O–0.55(0.5B$_2$O$_3$–0.5P$_2$O$_5$), the maximum ionic conductivity was achieved with 1.2×10^{-7} S/cm at room temperature. The ionic conductivity exhibited a maximum at $B_2O_3/(B_2O_3+P_2O_5)$=0.5 and the influence of added Li^+ ion differed depending on the $B_2O_3/(B_2O_3+P_2O_5)$ ratio.

In $B_2O_3/(B_2O_3+P_2O_5)$ ratio range <0.5, the ionic conductivity increased with increasing Li_2O concentration, and this is attributed to the increase of number of (B–O–P)$^-$groups. For $B_2O_3/(B_2O_3+P_2O_5)$ ratio >0.5, however, the ionic conductivity decreased with increasing Li_2O concentration due to the tendency of preferential association of Li^+ to P–O–P chains and the decrease of (B–O–P)$^-$ groups.

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